Learning from Additional Data under Bayesian Non-Identifiability: Two Neural Diagnostics

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Different parameter values can explain the same data

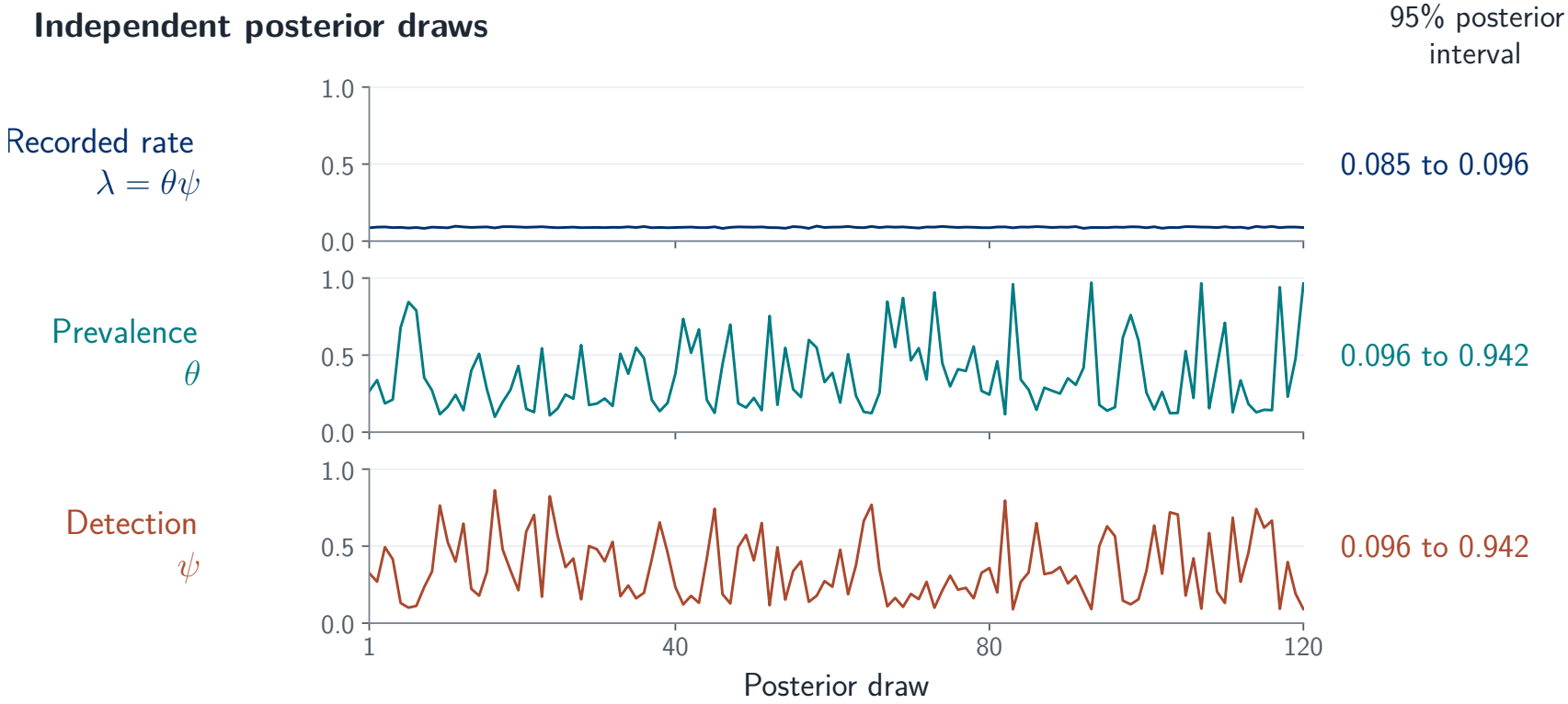
$$Y \sim \text{Binomial}(N, \theta\psi), \qquad \lambda = \theta\psi.$$

Prevalence θ	Detection probability ψ	Recorded rate λ
0.10	0.90	0.09
0.15	0.60	0.09
0.30	0.30	0.09

- Different combinations produce the **same distribution of observed data**.
- These observations cannot determine prevalence and detection separately.

Posterior traces in a non-identifiable model

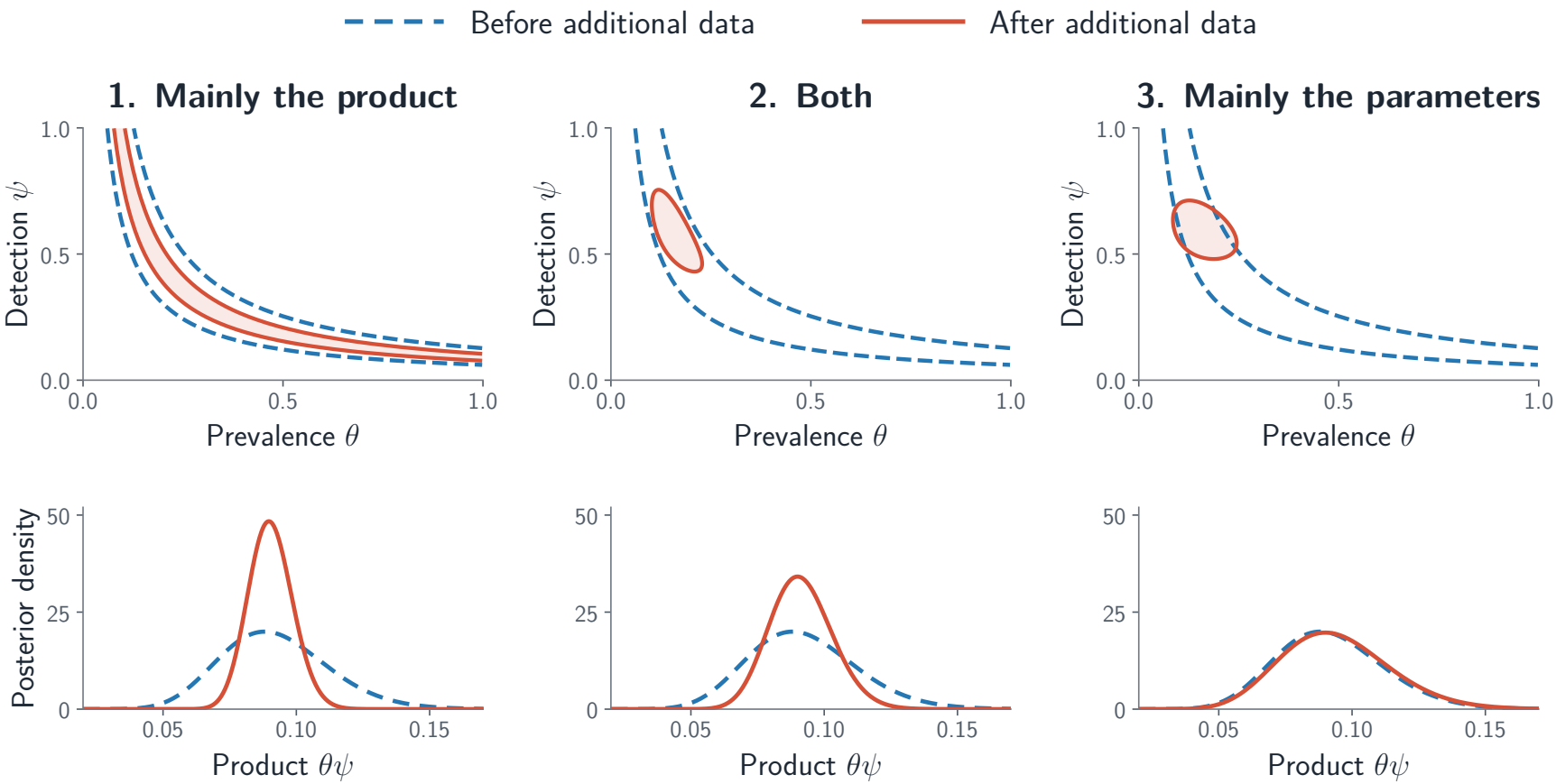
Illustration: 900 recorded cases out of 10,000 observations.
Prior: θ and ψ are independently uniform from 0 to 1.



These data estimate the product precisely, but cannot distinguish pairs with the same product.

Additional data can change the posterior in different ways

Each column adds a different dataset to the same baseline.



- **Posterior precision and identification**
  - Proper posteriors can exist without identification.
  - Prior dependence can persist with large samples.
  - Poirier (1998); Gustafson (2010); Moon and Schorfheide (2012).
- **Transparent parameterizations**
  - Separate identified and unidentified parts of causal models.
  - Expose prior dependence in the unidentified part.
  - Richardson, Evans, and Robins (2011).
- **Additional measurements**
  - Unknown test accuracy can leave prevalence unidentified.
  - Prior sensitivity may remain after adding tests.
  - Joseph, Gyorkos, and Coupal (1995); Dendukuri and Joseph (2001).

Two diagnostics for additional data

- **Fiber Contraction (FC): change in target uncertainty.**
  Compare conditional target variance before and after adding data.
- **Source-Partitioned Information Decomposition for Recoverability (SPIDR): allocation of information.**
  Distinguish the source of the KL update after adding data.
- **Computation from posterior samples.**
  Use neural networks for conditional moments and density ratios.

The data sources and the scientific target

In the product model, $\boldsymbol{\Theta}_{[2\times 1]} = (\theta, \psi)^\top$:

$$\underbrace{\lambda = \theta\psi}_{\text{fiber index}}, \qquad \underbrace{Z = \theta}_{\text{target: prevalence}} .$$

Composite data: recorded cases

$$Y_{\text{cmp}} \mid \boldsymbol{\Theta} \sim \text{Binomial}(N, \lambda).$$

Inform **only the product** λ. They cannot identify prevalence θ and detection ψ separately.

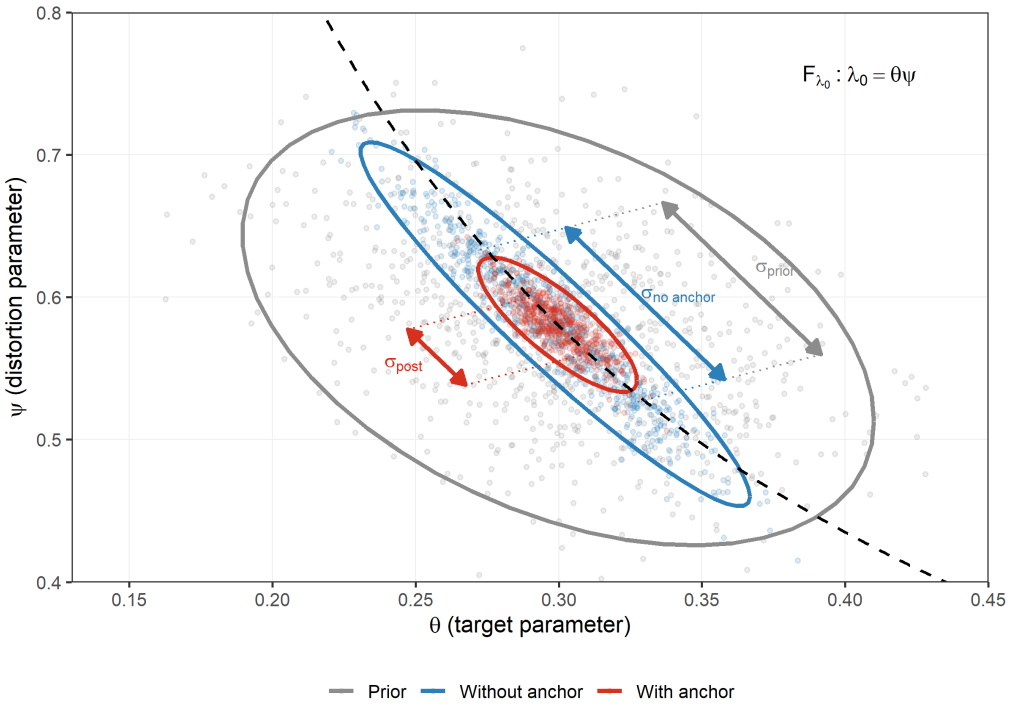
Anchor data: a detection study

$$Y_a \mid \boldsymbol{\Theta} \sim \text{Binomial}(K, \psi).$$

These data **directly inform the component** ψ, helping separate detection from prevalence.

In general: $\boldsymbol{\Theta}_{[d\times 1]}$, $\boldsymbol{\lambda}_{[q\times 1]} = \mathcal{H}(\boldsymbol{\Theta})$, and scalar $Z = g(\boldsymbol{\Theta})$.

Fiber, composite data, and anchor data



Schematic illustration; gray = prior

Fiber
Parameter combinations with the same $\lambda = \theta\psi$.

Composite posterior
Inform λ, but cannot distinguish points on a fiber.

Anchored posterior
Additional observations may help distinguish these points.

$$\mathcal{F}_\lambda = \{\boldsymbol{\Theta} : \mathcal{H}(\boldsymbol{\Theta}) = \lambda\}, \qquad \lambda : \text{ fiber index.}$$

Local FC: compare target variance at one fixed $\boldsymbol{\lambda}$.

$$\mathrm{FC}_{\mathrm{loc}}(\boldsymbol{\lambda}) = \frac{\mathrm{Var}(Z \mid \boldsymbol{\lambda}, y_{\mathrm{cmp}}, y_a)}{\mathrm{Var}(Z \mid \boldsymbol{\lambda}, y_{\mathrm{cmp}})}.$$

Integrated FC: average each variance, then take the ratio.

$$\mathrm{FC}_{\mathrm{int}} = \frac{\mathbb{E}_{\boldsymbol{\lambda}\sim p(\boldsymbol{\lambda}|y_{\mathrm{cmp}},y_a)}[\mathrm{Var}(Z \mid \boldsymbol{\lambda}, y_{\mathrm{cmp}}, y_a)]}{\mathbb{E}_{\boldsymbol{\lambda}\sim p(\boldsymbol{\lambda}|y_{\mathrm{cmp}},y_a)}[\mathrm{Var}(Z \mid \boldsymbol{\lambda}, y_{\mathrm{cmp}})]}.$$

Both averages use the **same anchored distribution** of $\boldsymbol{\lambda}$.

- $\mathrm{FC}_{\mathrm{int}} < 1$: Less uncertainty suggests better **practical identifiability**.

  Example: $\mathrm{FC}_{\mathrm{int}} = 0.2$: After adding the anchor data, average uncertainty about the target is **80% lower**, holding fixed what the composite data can identify.

- $\mathrm{FC}_{\mathrm{int}} \geq 1$: No average variance reduction. The anchor may still help.

FC computation: learn the mean, then the variance

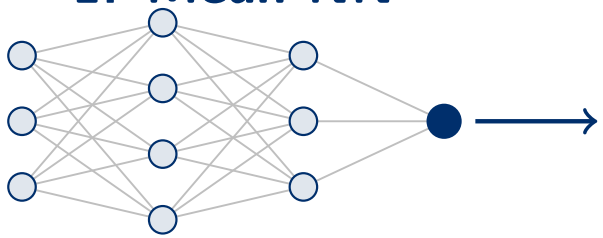
Fit this two-stage procedure **separately** using draws from $p(\boldsymbol{\Theta} \mid y_{\text{cmp}})$ and $p(\boldsymbol{\Theta} \mid y_{\text{cmp}}, y_a)$.

Residual-based variance estimation: Fan and Yao (1998).

Related ReLU estimators: Padilla et al. (2024).

Input **Neural network** **Output**

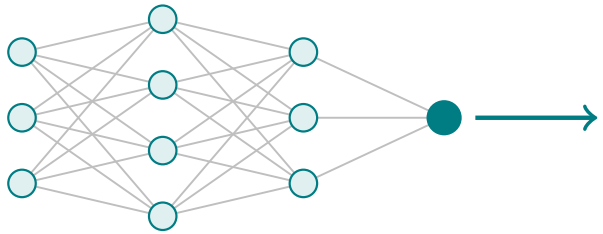
1. Mean NN

Fiber index $\boldsymbol{\lambda}_{[q\times 1]}$ →  → Conditional mean $\widehat{\mathbb{E}}[Z \mid \boldsymbol{\lambda}, \text{data}]$

Training response: Z

Freeze the mean NN; compute squared residuals.

2. Variance NN

Fiber index $\boldsymbol{\lambda}_{[q\times 1]}$ →  → Conditional variance $\widehat{\text{Var}}(Z \mid \boldsymbol{\lambda}, \text{data})$

Training response: $\left(Z - \widehat{\mathbb{E}}[Z \mid \boldsymbol{\lambda}, \text{data}]\right)^2$

Cross-fitting: train on other folds; predict the held-out fold.

- **FC measures changes in target uncertainty.**
  For a chosen Z, compare conditional variances before and after the anchor, at fixed $\boldsymbol{\lambda}$.

  Integrated FC < 1: lower conditional variance on average.

- **FC leaves the allocation of information unresolved.**
  A variance ratio cannot describe the full posterior update or separate changes across fibers from changes within fibers.

SPIDR: what share of the total posterior KL change occurs within fibers?

SPIDR: where does the posterior information go?

KL divergence measures the size of a distributional change.

$$D_{\text{total}} = D_{\text{KL}}\{p(\boldsymbol{\Theta} \mid y_{\text{cmp}}, y_a) \| p(\boldsymbol{\Theta} \mid y_{\text{cmp}})\},$$
$$D_{\text{across}} = D_{\text{KL}}\{p(\boldsymbol{\lambda} \mid y_{\text{cmp}}, y_a) \| p(\boldsymbol{\lambda} \mid y_{\text{cmp}})\},$$
$$D_{\text{within}} = \mathbb{E}_{\boldsymbol{\lambda} \sim p(\boldsymbol{\lambda} \mid y_{\text{cmp}}, y_a)}[D_{\text{KL}}\{p(\boldsymbol{\Theta} \mid \boldsymbol{\lambda}, y_{\text{cmp}}, y_a) \| p(\boldsymbol{\Theta} \mid \boldsymbol{\lambda}, y_{\text{cmp}})\}].$$

$$D_{\text{total}} = \underbrace{D_{\text{across}}}_{\text{change in fiber probabilities}} + \underbrace{D_{\text{within}}}_{\text{change at fixed fibers}}, \qquad \text{SPIDR} = \frac{D_{\text{within}}}{D_{\text{total}}}.$$

Example:

$$D_{\text{total}} = 2, \quad D_{\text{across}} = 0.4, \qquad \text{SPIDR} = \frac{2 - 0.4}{2} = 0.8.$$

80% of the new information from the anchor helps distinguish parameter values that the composite data cannot distinguish.

Original Θ: integrate out any parameters added only by the anchor.

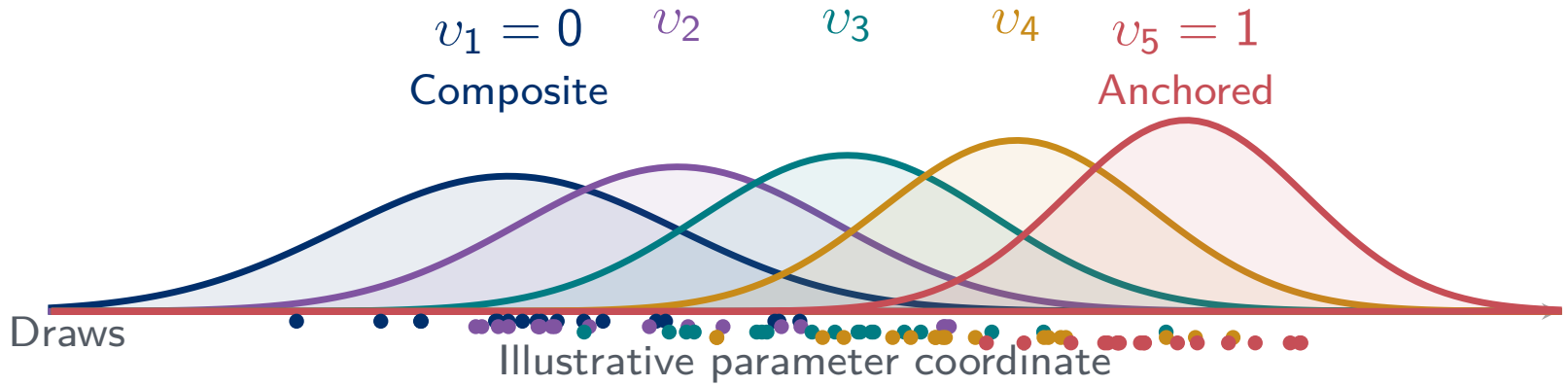
SPIDR computation: generate training samples

Sample the full model, including any anchor-only parameters $\eta_{[m\times 1]}$:

$$p_\upsilon(\Theta, \eta) \propto p(\Theta, \eta \mid y_{\text{cmp}})\, p(y_a \mid \Theta, \eta, y_{\text{cmp}})^\upsilon,$$

$$p(y_a \mid \Theta, \eta, y_{\text{cmp}}) = p(y_a \mid \Theta, \eta) \quad \big(Y_a \perp\!\!\!\perp Y_{\text{cmp}} \mid (\Theta, \eta)\big).$$

Illustration: $J = 5$ temperature classes, $0 = \upsilon_1 < \cdots < \upsilon_5 = 1$.



Illustrative Gaussian posteriors; one curve per temperature.

- Retain each draw's temperature label and original parameters Θ.
- Exclude η from NN inputs. Balance the temperature classes.

We adapt the auxiliary-distribution multiclass method of Srivastava et al. (2023) to posterior temperatures.

SPIDR computation: predict temperature probabilities

Cross-fitting: train on other folds; predict the held-out fold.

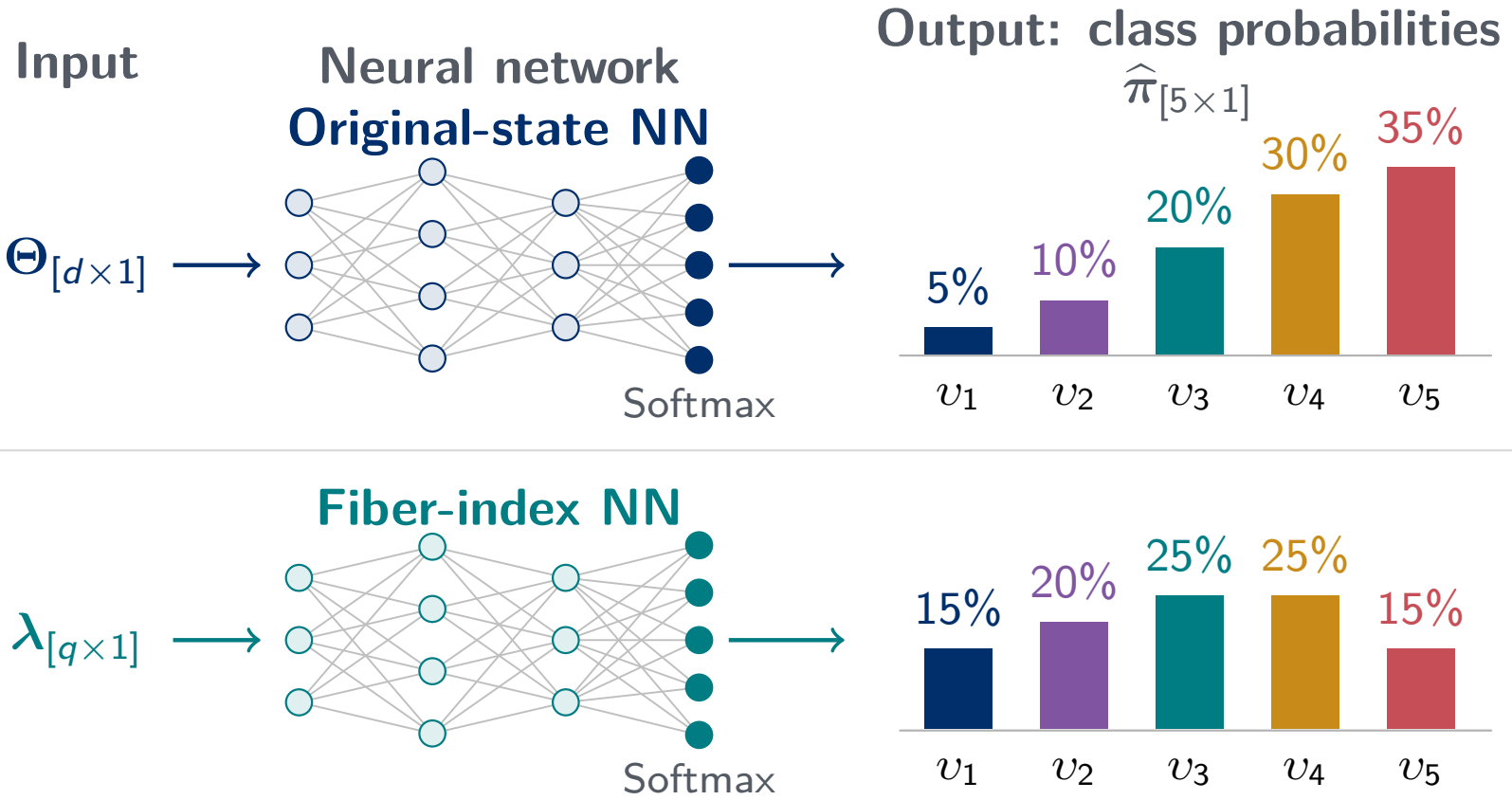


Illustration with $J = 5$ classes. Each set of probabilities sums to 100%.

Classifier density ratios (Mukherjee et al., 2020), with balanced classes:

$$\widehat{\mathcal{R}}(\text{input}) = \log \frac{\widehat{\Pr}(\upsilon = \upsilon_5 \mid \text{input})}{\widehat{\Pr}(\upsilon = \upsilon_1 \mid \text{input})}.$$

SPIDR computation: averaging NN predictions

Use the same anchored-posterior test draws for both NNs.

Total KL concerns the **original parameters**, after marginalizing any anchor-only parameters.

Average log ratios over test draws from all folds.

$$\widehat{D}_{\text{total}} = \text{mean}(\widehat{\mathcal{R}}_{\boldsymbol{\Theta}}), \qquad \widehat{D}_{\text{across}} = \text{mean}(\widehat{\mathcal{R}}_{\boldsymbol{\lambda}}).$$

Subtract to get within-fiber information.

$$\widehat{D}_{\text{within}} = \widehat{D}_{\text{total}} - \widehat{D}_{\text{across}}.$$

Divide by the total to get SPIDR.

$$\widehat{\text{SPIDR}} = \frac{\widehat{D}_{\text{within}}}{\widehat{D}_{\text{total}}}.$$

FC and SPIDR answer complementary questions

	Low SPIDR	**High SPIDR**
Small FC	Target variance contracts. Most KL learning occurs across fibers.	Target variance contracts. Most KL learning occurs within fibers.
FC near / above 1	Limited target precision gain. Most KL learning occurs across fibers.	Limited target precision gain. Learning may concern other parameters, location, or shape.

High SPIDR alone does not establish target recovery or point identification.

Simulation: main findings

Product model: $\lambda = \theta\psi$. Target: $Z = \psi$.

$$Y_{\text{cmp}} \sim \text{Binomial}(200, \lambda), \qquad Y_a \sim \text{Binomial}(K, \psi).$$

- **Larger anchors improve target precision.**
  FC decreases and within-fiber KL increases, while SPIDR remains near one.
- **What we measure matters.**
  With the same added-data budget, more direct measurements of ψ produce stronger contraction and a larger within-fiber share.
- **The prior still matters.**
  The tested likelihood families give similar diagnostics, but changing the prior affects contraction and interval coverage even when SPIDR is high.

200 replicates per condition. Detailed results are in the appendix.

BEACON: prevalence and recorded cases

- 2023 American Indian/Alaska Native (AI/AN) births in Oklahoma.
- Oklahoma State Department of Health (OSDH): counts in 28 of 77 counties.

$$Y_c^{\mathrm{OSDH}} \sim \mathrm{Binomial}(n_c^{\mathrm{OSDH}}, \rho_c \phi_c).$$

- Y_c^{OSDH}: **number of gestational diabetes cases** recorded by OSDH in county c.
- n_c^{OSDH}: **total number of AI/AN births** in county c (known).
- ρ_c: true prevalence; ϕ_c: unknown recording probability.
- Likelihood depends only on $\lambda_c = \rho_c \phi_c$.

$$\text{Target: } Z_c = \rho_c, \qquad \boldsymbol{\lambda}_{[77\times 1]} = (\rho_c \phi_c)_{c=1}^{77}.$$

Diagnostics condition on all 77 products to define a fiber; only 28 observed products are identifiable.

Prevalence and OSDH recording each have a county hierarchy:

$$\text{logit}(\rho_c) = \beta_0 + \mathbf{x}_c^\top \boldsymbol{\beta} + u_c^{(\rho)} + v_c^{(\rho)},$$
$$\text{logit}(\phi_c) = \gamma_0 + \gamma_1 \log(x_c^{\text{AIAN}}) + u_c^{(\phi)} + v_c^{(\phi)}.$$

$\mathbf{x}_{c[3\times 1]}$: SNAP participation, low food access, prenatal care.

x_c^{AIAN}: county AI/AN population proportion.

For each process $s \in \{\rho, \phi\}$:

$$\mathbf{u}^{(s)}_{[77\times 1]} \sim \text{ICAR}(G, \sigma_{u,s}), \quad \sum_{c=1}^{77} u_c^{(s)} = 0, \qquad v_c^{(s)} \overset{\text{iid}}{\sim} N(0, {\sigma_{v,s}}^2).$$

The prior connects counties; the OSDH likelihood still sees only $\rho_c \phi_c$.

Cherokee Nation (CN) supplies usable counts in 12 counties; 11 also have OSDH observations. Both sources share prevalence ρ_c:

$$Y_c^{\mathrm{CN}} \sim \mathrm{Binomial}(n_c^{\mathrm{CN}}, \rho_c \kappa_c).$$

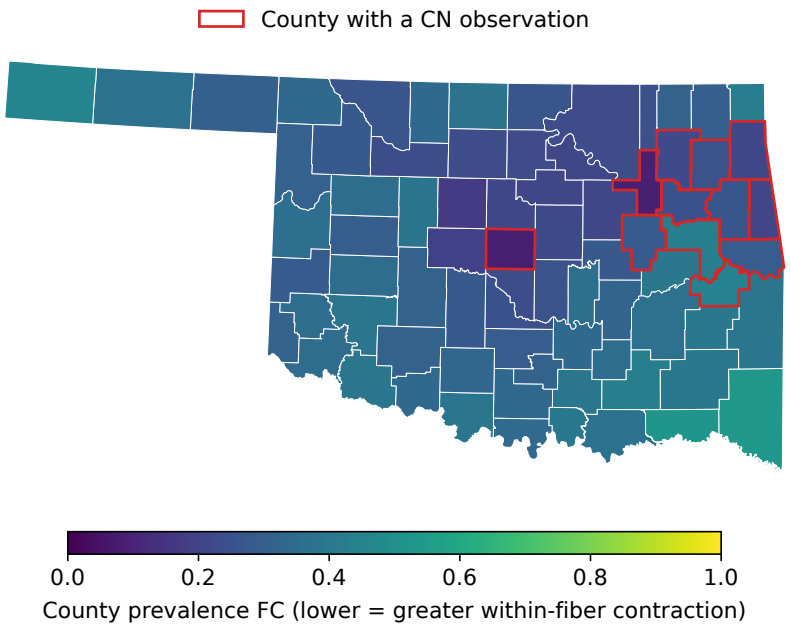
κ_c: effective CN coverage and case capture, relative to the county birth total.

$$\mathrm{logit}(\kappa_c) = \alpha_0 + \alpha_1 \,\mathrm{clinics}_c + \alpha_3 \,\mathrm{spillover}_c,$$

$$\mathrm{spillover}_c = \frac{1}{|\mathcal{N}(c)|} \sum_{j \in \mathcal{N}(c)} \mathrm{clinics}_j.$$

- $\mathrm{clinics}_c$: IHS-listed facilities in county c;
- $\mathcal{N}(c)$: adjacent Oklahoma counties.

BEACON: FC and SPIDR



Prevalence FC: median 0.311 Range: 0.083–0.534

SPIDR = 0.7087

- **FC:** all 77 counties gain precision; the median reduction in averaged conditional prevalence variance is **68.9%**.
- **SPIDR:** 70.9% of the original-parameter KL update occurs within fibers, after marginalizing CN capture coefficients.

Proxy SVAR: observed series and structural shocks

Model: Caldara and Herbst (2019). At month t, $\mathbf{y}_{t[4\times1]}$ contains interest rates, production, unemployment, and prices. $\mathbf{x}_{t[49\times1]}$ contains 12 lags and an intercept.

The structural model

$$\mathbf{A}_0{}^{\top}\mathbf{y}_t = \mathbf{A}_+{}^{\top}\mathbf{x}_t + \mathbf{e}_t, \qquad \mathbf{e}_t \sim N(\mathbf{0}, \mathbf{I}_4).$$

$\mathbf{A}_{0[4\times4]}$: invertible structural matrix; $\mathbf{A}_{+[49\times4]}$: lag/intercept coefficients. $\mathbf{e}_{t[4\times1]}$: four latent, independent, unit-variance shocks. $\mathbf{I}_4$ is the identity matrix.

The same model as a regression

$$\mathbf{y}_t = \mathbf{B}^{\top}\mathbf{x}_t + \varepsilon_t, \qquad \mathbf{B}_{[49\times4]} = \mathbf{A}_+\mathbf{A}_0{}^{-1}.$$

$\varepsilon_{t[4\times1]} = \mathbf{y}_t - \mathbf{B}^{\top}\mathbf{x}_t$ is the residual in the observed series.

$$\boxed{\varepsilon_t = \mathbf{A}_0{}^{-\top}\mathbf{e}_t} \qquad \Longleftrightarrow \qquad \boxed{\mathbf{e}_t = \mathbf{A}_0{}^{\top}\varepsilon_t}.$$

The unknown matrix $\mathbf{A}_0$ links residuals to structural shocks.

The macro-data likelihood depends only on $(\mathbf{B}, \Sigma)$:

$$\varepsilon_t \sim N(\mathbf{0}, \Sigma), \qquad \Sigma_{[4\times 4]} = (\mathbf{A}_0 \mathbf{A}_0{}^\top)^{-1}.$$

For a rotation $\mathbf{O}_{[4\times 4]}$ with $\mathbf{O}^\top \mathbf{O} = \mathbf{I}_4$,

$$(\mathbf{A}_0, \mathbf{A}_+) \longmapsto (\mathbf{A}_0 \mathbf{O}, \mathbf{A}_+ \mathbf{O})$$

leaves $(\mathbf{B}, \Sigma)$ unchanged, so macro data cannot distinguish points on the same fiber.

Anchor: an observed policy proxy measures the first shock.

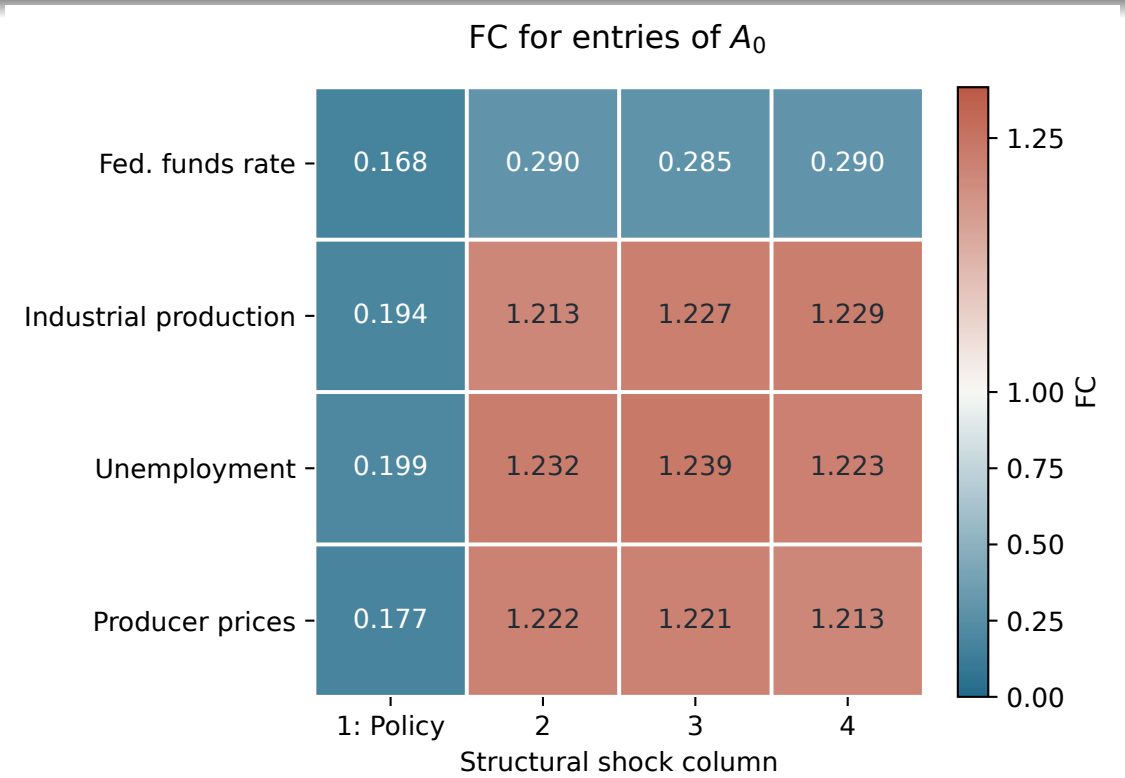
$$z_t = b\, e_{1t} + \sigma_\nu \nu_t, \qquad \nu_t \sim N(0,1), \quad \nu_t \perp \mathbf{e}_t.$$

$$\mathbf{e}_t = \mathbf{A}_0{}^\top \varepsilon_t = \mathbf{A}_0{}^\top \left(\mathbf{y}_t - \mathbf{B}^\top \mathbf{x}_t\right).$$

- z_t: observed proxy; e_{1t}: latent policy shock, computed from the data at each parameter draw.
- b: unknown loading; $\sigma_\nu > 0$: unknown proxy-noise scale.

With $b \neq 0$, the proxy informs the first shock direction; the other three can still rotate.

Proxy SVAR: FC and SPIDR



$$\text{SPIDR} = 0.8042$$

- **FC:** averaged conditional variances fall by 80–83% in the policy column, but increase in the lower-right block.
- **SPIDR:** 80.4% of the KL update to $(\mathbf{A}_0, \mathbf{A}_+)$ is within fibers, after marginalizing proxy loading and noise.
- The proxy informs the policy shock, but rotations among the other shocks remain unresolved.

Geese: learning diet from isotope measurements

Brent geese eat four food sources. We estimate **their dietary proportions** from plasma carbon isotope measurements using the model of Parnell et al. (2013).

Eight sampling periods: fit each period independently, with its own diet and carbon residual scale. Periods share the fixed food-source inputs.

y_{iC}: carbon isotope measurement in sample $i = 1, \ldots, n_g$ of period g. w_k: proportion of food $k = 1, \ldots, 4$, with $\mathbf{w}_{[4\times 1]}$, $w_k > 0$, and $\sum_k w_k = 1$.

Fixed food inputs: ζ_{kC} is carbon concentration; d_{kC} and v_{kC} are carbon isotope means and variances corrected for changes from food to tissue.

$$\underbrace{W_{kC}(\mathbf{w}) = \frac{w_k \zeta_{kC}}{\sum_{\ell=1}^{4} w_\ell \zeta_{\ell C}}}_{\text{concentration-adjusted proportion}}, \qquad \underbrace{\mu_C(\mathbf{w}) = \sum_{k=1}^{4} W_{kC}(\mathbf{w}) d_{kC}}_{\text{predicted mean}}.$$

$$V_C(\mathbf{w}, \sigma_C) = \sum_{k=1}^{4} W_{kC}(\mathbf{w})^2 v_{kC} + {\sigma_C}^2, \qquad y_{iC} \mid \mathbf{w}, \sigma_C \sim N(\mu_C, V_C).$$

V_C: total variance; $\sigma_C > 0.001$: unknown residual standard deviation. Observations are conditionally independent.

Carbon data inform only two combinations of the unknowns:

$$\boldsymbol{\lambda}_{[2\times 1]} = \left(\mu_C(\mathbf{w}), V_C(\mathbf{w}, \sigma_C)\right)^\top.$$

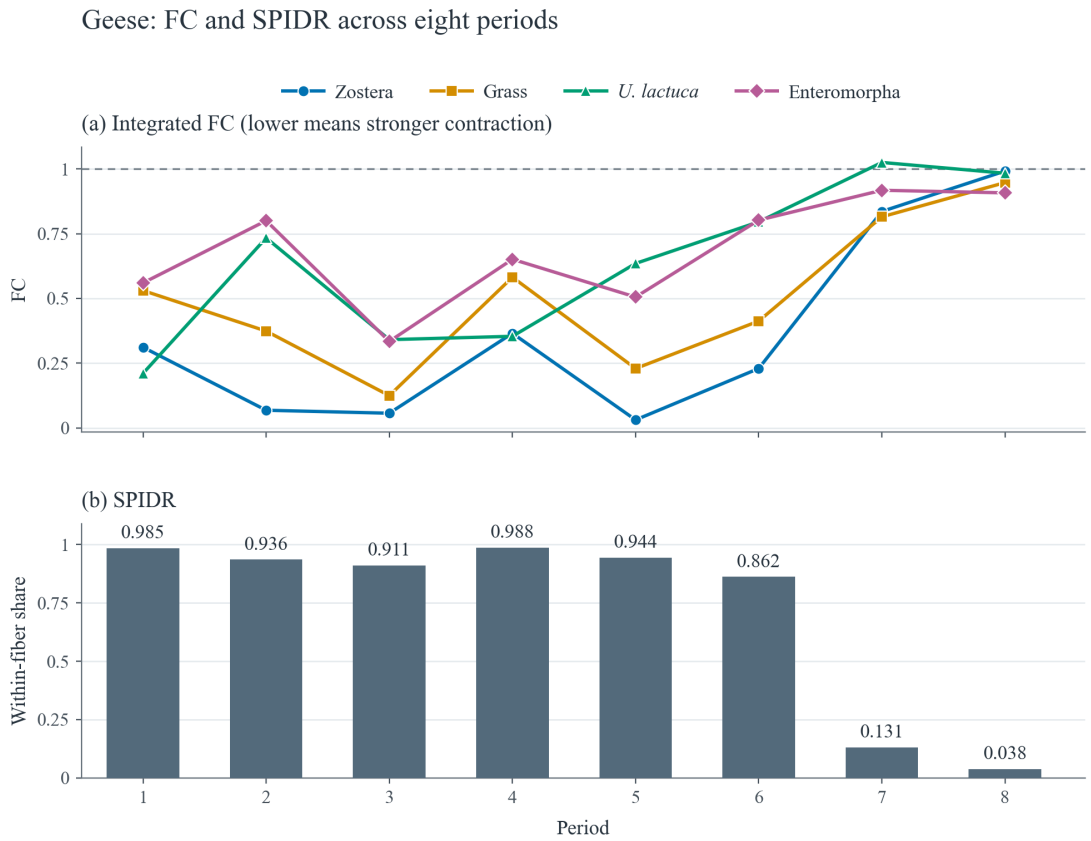
Four food proportions sum to one, leaving **three free weights**. Together with the unknown carbon residual variance, the carbon model has **four free parameters**, but carbon data identify only **two combinations**. Fixing them defines a fiber.

Anchor: nitrogen from the same samples. Apply the same formulas with C replaced by N. Nitrogen adds information about $\mu_N(\mathbf{w})$ and $V_N(\mathbf{w}, \sigma_N)$, using the same diet but different source inputs. For SPIDR, retain $(\mathbf{w}, \sigma_C)$ and integrate out the new scale σ_N.

Nitrogen adds constraints, but need not uniquely determine all four dietary proportions.

Geese: FC and SPIDR across eight periods

Eight periods over two winters, fitted **independently**.



- **Periods 1–6:** diet FCs below one, high SPIDR. Greater dietary precision, mostly within-fiber learning.
- **Periods 7–8:** FCs nearer one, low SPIDR. Smaller precision gains, mostly across-fiber learning.

Predator interference: experiment and model

Predatory bugs (*M. pygmaeus*) eat moth eggs, but can interfere with one another. We use the 12-hour trials of Papanikolaou et al. (2021) to separate attack rate from interference.

$$Y_i \mid \boldsymbol{\Theta} \sim \text{Binomial}(N_{0i}, \pi_i), \qquad a_{\text{eff},i} = \frac{a}{1 + \chi(n_i - 1)}.$$

$$-\log(1 - \pi_i) + a_{\text{eff},i}\tau N_{0i}\pi_i = a_{\text{eff},i} n_i.$$

Observed or known for trial i:

- N_{0i}: initial egg count; n_i: number of predators; Y_i: eggs consumed.

Unknown parameters: $\boldsymbol{\Theta}_{[3\times 1]} = (a, \tau, \chi)^\top$.

- a: attack rate without interference.
- τ: handling time per egg.
- χ: interference strength between predators.

Calculated quantities: $a_{\text{eff},i}$ is the attack rate after interference; π_i is the consumption probability, solved from the second equation.

Composite data: 59 trials with two predators ($n_i = 2$). Their likelihood depends only on

$$\boldsymbol{\lambda}_{[2\times 1]} = \left(\frac{a}{1+\chi}, \tau\right)^\top = (a_{\text{eff}}, \tau)^\top.$$

Five initial prey densities distinguish a_{eff} and τ. **Attack rate a and interference χ are not separately identifiable.** Same fiber: at fixed τ, both $(a, \chi) = (0.4, 1)$ and $(0.6, 2)$ give $a_{\text{eff}} = 0.2$, hence the same composite-data distribution.

Anchor data: 79 additional trials with one predator ($n_i = 1$).

$$a_{\text{eff},i} = \frac{a}{1+\chi(1-1)} = a.$$

Learning a separately helps distinguish attack from interference.

Predator interference: results for interference strength

Target: $Z = \chi$	
Diagnostic	Estimate
FC	0.0488
SPIDR	0.6455

- **Target precision:** the averaged conditional variance of interference χ falls by about **95.1%**.
- **Information allocation: 64.5%** of the posterior KL update is within fibers; the rest changes the posterior of $\boldsymbol{\lambda}$.
- Single-predator trials help distinguish the attack and interference parameters that two-predator trials cannot separate.

Main takeaways

- **FC measures changes in target uncertainty.**
  FC < 1 suggests better practical identifiability, holding fixed what the composite data can identify.
- **SPIDR shows where the anchor's information goes.**
  It measures the share that helps distinguish parameter values the composite data cannot distinguish.
- **Interpret FC, SPIDR, and total KL together.**
  High SPIDR does not guarantee a large update or a precise target.
- **NNs estimate both diagnostics from posterior samples.**
  FC uses regression, while SPIDR uses classification.

Thank you

FC: what its value tells us

- **No change within fibers: FC = 1.**
  If the added likelihood depends only on $\boldsymbol{\lambda}$, the conditional distribution of Z stays unchanged.

- **Target concentration: FC approaches 0.**
  This holds when the target concentrates in mean square and the baseline average variance stays bounded away from zero.

- **FC can exceed 1.**
  A particular dataset can increase conditional uncertainty.

FC describes the chosen target, not every parameter.

- **SPIDR = 0:** all information is across fibers; the conditional posterior within each fiber is unchanged.
- **SPIDR = 1:** all information is within fibers; the posterior distribution of $\boldsymbol{\lambda}$ is unchanged.
- **Coordinate invariance:** a one-to-one relabeling of the same state and fibers preserves SPIDR.

A high share need not mean a large update or precise target estimates.

With finite KL and $D_{\text{total}} > 0$, SPIDR lies in $[0, 1]$.
Report the absolute KL values too.

$$p(\Theta) := p(\Theta \mid y_{\text{cmp}}), \qquad q(\Theta) := p(\Theta \mid y_{\text{cmp}}, y_a).$$

Let p_λ, q_λ be their marginals for $\lambda = \mathcal{H}(\Theta)$.

Log density-ratio decomposition:

$$\log \frac{q(\Theta)}{p(\Theta)} = \log \frac{q_\lambda(\lambda)}{p_\lambda(\lambda)} + \log \frac{q(\Theta|\lambda)}{p(\Theta|\lambda)}.$$

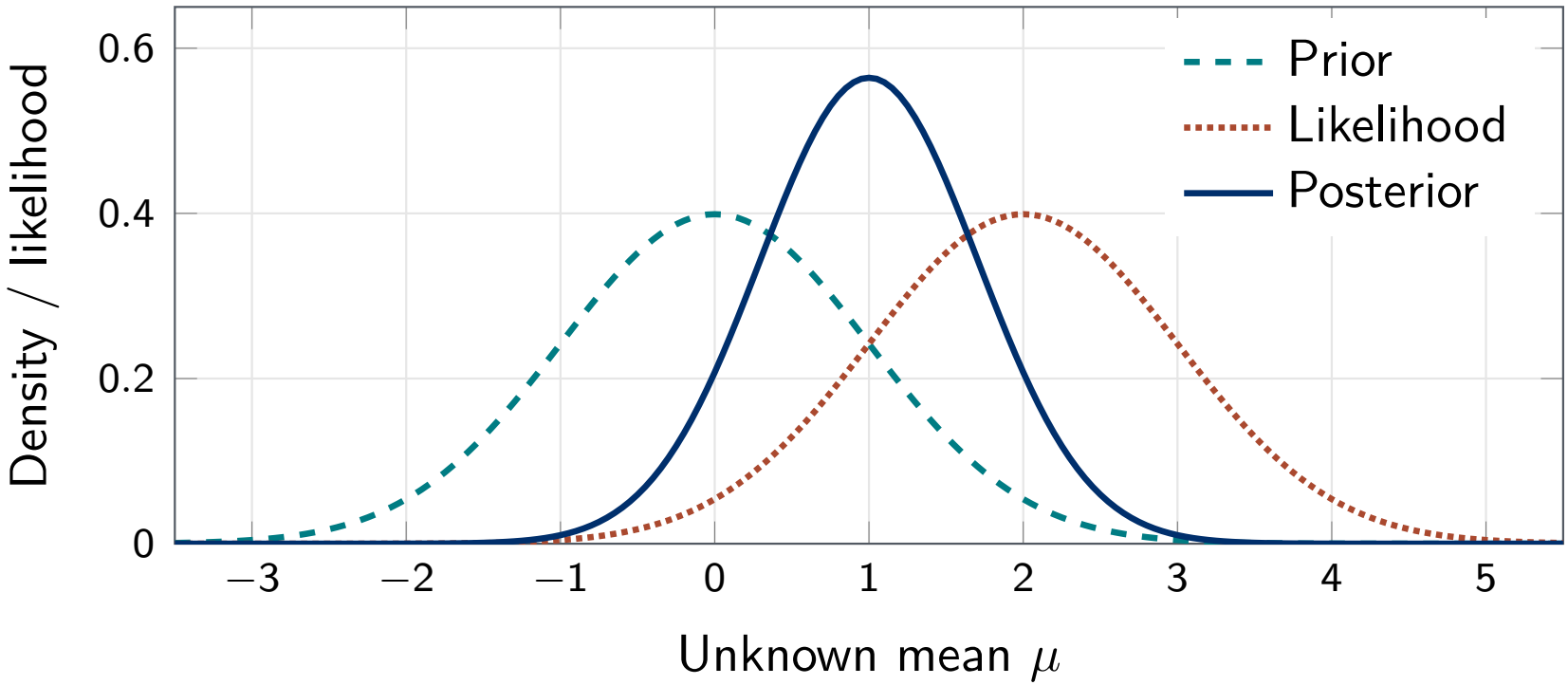
Iterated expectation under the anchored posterior q:

$$\begin{aligned} D_{\text{total}} &= \mathbb{E}_q\Big[\log \frac{q(\Theta)}{p(\Theta)}\Big] \\ &= \mathbb{E}_{q_\lambda}\Big[\log \frac{q_\lambda(\lambda)}{p_\lambda(\lambda)} + \mathbb{E}_{q(\Theta|\lambda)}\Big[\log \frac{q(\Theta|\lambda)}{p(\Theta|\lambda)}\Big]\Big] \\ &= \underbrace{D_{\text{KL}}(q_\lambda \| p_\lambda)}_{D_{\text{across}}} + \underbrace{\mathbb{E}_{q_\lambda}\Big[D_{\text{KL}}\{q(\Theta \mid \lambda) \| p(\Theta \mid \lambda)\}\Big]}_{D_{\text{within}}} \\ &= D_{\text{across}} + D_{\text{within}}. \end{aligned}$$

At fixed λ, the across log ratio is constant in Θ. Only the within term retains an inner expectation.

Conditional densities use a common reference measure on each fiber.

$$\underbrace{p(\mu \mid y)}_{\text{posterior}} \propto \underbrace{p(y \mid \mu)}_{\text{likelihood}} \underbrace{p(\mu)}_{\text{prior}}.$$



$$Y \mid \mu \sim N(\mu, 1), \qquad y = 2,$$

$$\mu \sim N(0, 1), \qquad \mu \mid y = 2 \sim N\left(1, \frac{1}{2}\right).$$

A precise posterior does not guarantee identification

- **Poirier (1998):** A proper prior can yield a posterior even when some parameters are **not identified**.
- **Gustafson (2010):** Even with large samples, uncertainty can remain and **depend on the prior**.
- **Moon and Schorfheide (2012):** Credible sets can exclude values that the **data alone cannot rule out**.

Posterior precision alone does not establish identification.

- **Richardson, Evans, and Robins (2011):** Transparent parameterization separates identified quantities from those that still depend on the conditional prior.

Our product example: use (λ, ψ) with $\lambda = \theta\psi$.

$$\underbrace{p(\psi \mid \lambda, y_{\text{cmp}})}_{\text{conditional posterior}} = \underbrace{p(\psi \mid \lambda)}_{\text{conditional prior}} .$$

At fixed λ, these data do not update uncertainty about ψ.

Reparameterization shows what the data identify and what still depends on the prior.

Additional measurements can still leave parameters unidentified

- **Joseph, Gyorkos, and Coupal (1995):** Joint Bayesian inference for prevalence and test accuracy **without a gold-standard test**.

  With two tests in one population, there are **3 free outcome probabilities but 5 unknowns**: prevalence and each test's sensitivity and specificity, even assuming conditional independence.

- **Dendukuri and Joseph (2001):** Allowed dependence between tests. In their nonidentified settings, **prior influence remained even with large samples**.

Additional measurements can help without identifying every parameter.

FC computation: from NN predictions to FC

Both fitted variance NNs use the **same held-out inputs**:

$$\boldsymbol{\lambda}^{(s)} \sim p(\boldsymbol{\lambda} \mid y_{\text{cmp}}, y_a).$$

Illustration: variance predictions and local FC

Input	Composite variance	Anchored variance	Local FC
$\boldsymbol{\lambda}^{(1)}$	0.020	0.010	0.50
$\boldsymbol{\lambda}^{(2)}$	0.040	0.008	0.20
$\boldsymbol{\lambda}^{(3)}$	0.060	0.012	0.20
Average	**0.040**	**0.010**	

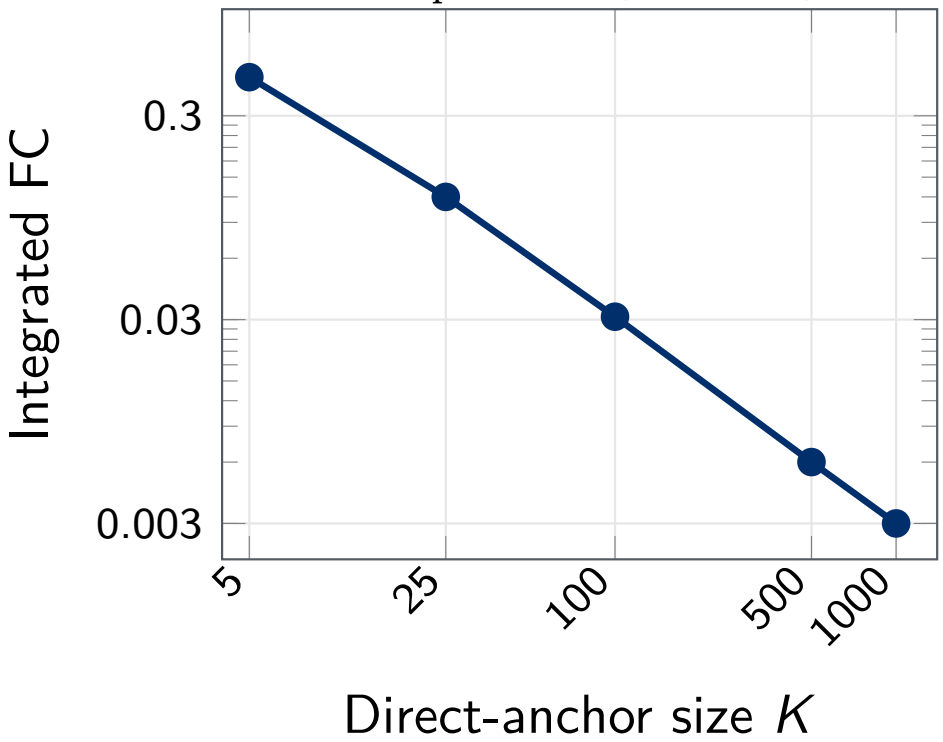
$$\widehat{\text{FC}_{\text{int}}} = \frac{\text{composite + anchor average}}{\text{composite-only average}} = \frac{0.010}{0.040} = 0.25.$$

Pool held-out predictions from all folds before taking the ratio.

In this example, averaged conditional variance falls by 75%.

Anchor size changes contraction far more than the share

200 replicates. $Y_{\text{cmp}} \sim \text{Bin}(200, \theta\psi)$; $Y_a \sim \text{Bin}(K, \psi)$; target $Z = \psi$.



K	Within KL	SPIDR
5	0.710	0.993
25	1.494	0.997
100	2.190	0.998
500	2.983	0.998
1000	3.313	0.998

$K = 0$: FC $= 1$, all KLs $= 0$, SPIDR undefined.

A nearly constant SPIDR can accompany very different precision gains.

Composite-data likelihood, with $K = 100$

Likelihood	FC	SPIDR
Binomial	0.031	0.998
Poisson	0.031	0.998
Gaussian	0.030	0.995

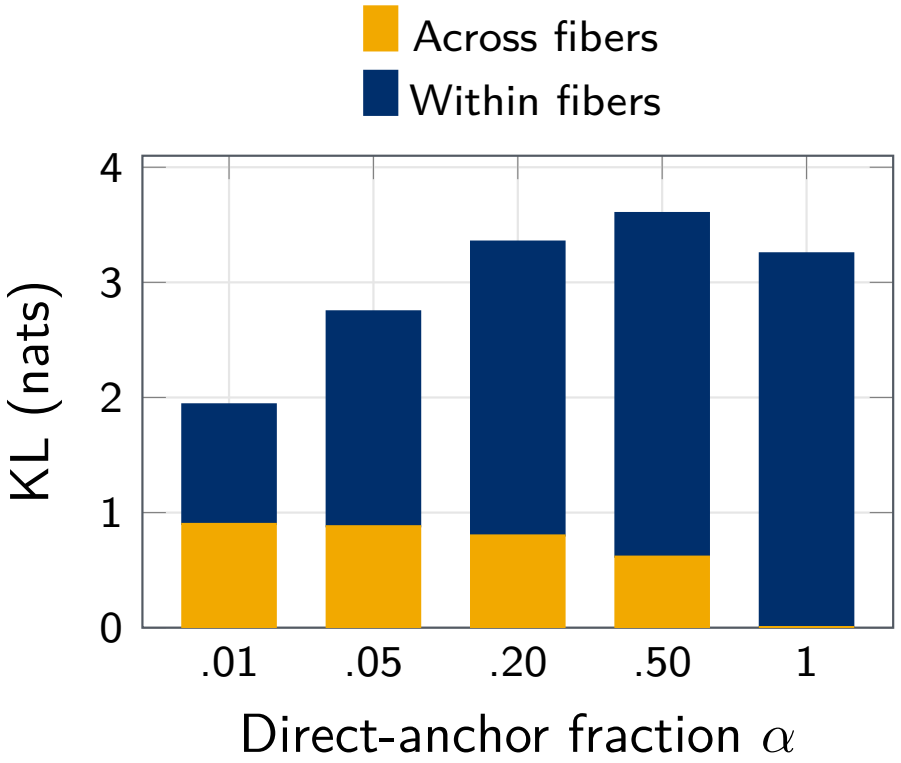
Prior sensitivity, with $K = 100$

Prior	FC	SPIDR	95% coverage of ψ
Flat	0.030	0.998	0.955
Near-centered, moderate	0.155	0.997	0.965
Near-centered, concentrated	0.462	0.981	0.785
Miscentered, moderate	0.171	0.937	0.885

High SPIDR does not guarantee frequentist coverage.

The same data budget can learn different things

1,000 added trials: M measure λ, K measure ψ, and $\alpha = K/1000$.



α	FC	SPIDR
0.01	0.273	0.549
0.05	0.061	0.701
0.20	0.015	0.778
0.50	0.006	0.843
1.00	0.003	0.998

More direct measurements improve target contraction and increase the within-fiber share.